

## STAT 341: Tutorial Test 2 - Sample ~~1~~<sup>2</sup>

First name .....

Last name .....

Signature .....

ID Number .....

Email .....

**Time allowed: 50 minutes**

### Special instructions

- All numeric answers should be written either as a number to three decimal places or as a fraction reduced to its lowest terms.
- If working is not specifically requested, a correct answer will receive full marks. However, an incorrect answer could still receive credit if working is provided. An incorrect answer with no working will receive zero marks.

**Please read:** this practice test is designed to give you a clear sense of the sort of things that can come up in exams. Please note that this is, of course, only a subset of the material that could be covered. I would **strongly** encourage you to attempt this test under exam conditions - do not be tempted to look up something in your notes until afterwards! This is one reason why it is common for students to find real exams much harder than practice ones, even though the material is of an equivalent difficulty!

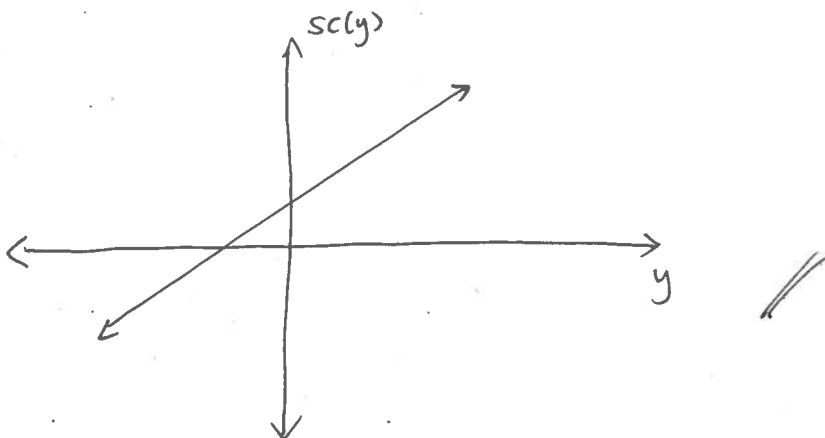
1. [5 marks] Derive and draw the sensitivity curve for the Horvitz-Thompson estimator, where  $y_u$  has an inclusion probability of  $\pi_u > 0$ .

Let  $S = \{y_1, \dots, y_n\}$  be a sample, and assume that the additional observation  $y$  has non-zero inclusion prob.

Denote  $\pi_1, \dots, \pi_n$  as inclusion prob. for observations in  $S$ .

Denote  $\pi_1^*, \dots, \pi_n^*, \pi^*$  as inclusion prob. for observations in  $S \cup \{y\}$ , since inclusion of  $y$  can change the inclusion prob. of other observations.

$$\begin{aligned}
 SC(y) &= (n+1)(a_{HT}(S \cup \{y\}) - a_{HT}(S)) \\
 &= (n+1)\left(\sum_{u \in S \cup \{y\}} \frac{y_u}{\pi_u^*} - \sum_{u \in S} \frac{y_u}{\pi_u}\right) \\
 &= (n+1)\left(\sum_{u \in S} \frac{y_u}{\pi_u^*} + \frac{y}{\pi^*} - \sum_{u \in S} \frac{y_u}{\pi_u}\right) \\
 &= (n+1) \sum_{u \in S} \left(\frac{1}{\pi_u^*} - \frac{1}{\pi_u}\right) y_u + \frac{(n+1)}{\pi^*} y \quad (\text{an affine function of } y)
 \end{aligned}$$



2. [4 marks] Based on a given sampling design, consider taking a sample of size  $n$ , without replacement, from a population  $\mathcal{P}$  in which the probability of inclusion and the joint probability of inclusion are  $\pi_u$  and  $\pi_{uv}$ , respectively. Show that  $\sum_{v \in \mathcal{P}} \pi_{uv} = n\pi_u$ .  
You can use the result, without proof, that the sum of probability of inclusion over the population is equal to the sample size.

Define the indicator  $I(x) = \begin{cases} 1 & \text{if } x \text{ in sample} \\ 0 & \text{else.} \end{cases}$

$$\begin{aligned} \sum_{v \in \mathcal{P}} \pi_{uv} &= \sum_{v \in \mathcal{P}} P(I(v)=1 \text{ and } I(u)=1) \\ &= \sum_{v \in \mathcal{P}} P(I(v)=1 | I(u)=1) \pi_u \quad (*) \end{aligned}$$

Note that  $P(I(v)=1 | I(u)=1)$  is the probability of selecting unit  $v$  ( $\neq u$ ) from the population  $\mathcal{P}$  after unit  $u$  is selected. This population is exactly  $\mathcal{P} \setminus \{u\}$ , and there are  $n-1$  units to select.

$$\begin{aligned} \Rightarrow (*) &= \pi_u \left( \sum_{v \in \mathcal{P} \setminus \{u\}} \overset{\pi_v}{P(I(v)=1)} + 1 \right) \\ &= \pi_u ((n-1) + 1) \\ &= n\pi_u. \quad \square \end{aligned}$$

3. [2 marks] The variance of the Horvitz-Thompson estimator  $a_{HT}(\mathcal{S})$  is

$$\text{Var}(a_{HT}(\mathcal{S})) = \sum_{u \in \mathcal{P}} \sum_{v \in \mathcal{P}} (\pi_{uv} - \pi_u \pi_v) \frac{y_u}{\pi_u} \frac{y_v}{\pi_v}$$

Write down the Horvitz-Thompson estimate of the variance based on the sample  $\mathcal{S}$ .

We notice that  $\text{Var}(a_{HT}(\mathcal{S}))$  is a total quantity.

$$\therefore \hat{\text{Var}}(a_{HT}(\mathcal{S})) = \sum_{u \in \mathcal{S}} \sum_{v \in \mathcal{S}} [(\pi_{uv} - \pi_u \pi_v) \frac{y_u}{\pi_u} \cdot \frac{y_v}{\pi_v}] \div \pi_{uv} //$$

4. Consider a population of size  $N = 100$  from which we take a random sample of size  $n = 4$ . The inclusion probabilities and the observed values  $y_i$  from the sample are provided in the table below.

|         |      |      |      |      |
|---------|------|------|------|------|
| $y_i$   | 1    | 2    | 4    | 6    |
| $\pi_i$ | 1/25 | 2/25 | 2/25 | 4/25 |

- (a) [3 marks] Calculate the HT estimate for the population's proportion of units which are less than or equal to 2.

$$P(Y \leq y) = \sum_{u \in P} I(y_u \leq 2) / N, \text{ where } N = 100.$$

We notice that proportion is an average of indicators, so

$$\begin{aligned} a_{HT}(s) &= \frac{1}{N} \sum_{u \in S} I(y_u \leq 2) / \pi_u \\ &= \frac{1}{100} \left( \frac{1}{(1/25)} + \frac{1}{(2/25)} \right) = \frac{3}{8} \quad // \end{aligned}$$

- (b) [3 marks] Calculate the HT estimate of the population's median.

We compute the HT estimates of  $P(Y \leq y) = F(y)$ ,

and we note that  $\hat{F}_{HT}(2) = \frac{3}{8} < \frac{1}{2}$  from part a).

$$\text{For } y \in [4, 6), \hat{F}_{HT}(y) = \frac{1}{100} \left( \frac{1}{(1/25)} + \frac{1}{(2/25)} + \frac{1}{(2/25)} \right) = \frac{1}{2}$$

For  $y = 6$ ,  $\hat{F}_{HT}(y) > \frac{1}{2}$ , since  $\pi_6 > 0$ .

Since  $\text{median} = \inf \{ y_u : F(y_u) \geq \frac{1}{2}, u \in P \}$ ,

$$\text{median} = \inf [4, 6) = 4.$$

However, any number in  $[4, 6)$  can be a HT estimate of median based on this sample. //

5. Consider the population  $\mathcal{P} = \{1, 3, 5\}$  from which a random sample of size 2 is selected. The sampling design and all possible samples are provided below.

| $S$    | $\{1, 3\}$ | $\{1, 5\}$ | $\{3, 5\}$ |
|--------|------------|------------|------------|
| $P(S)$ | 0.2        | 0.5        | 0.3        |

- (a) [3 marks] Calculate the sampling bias of the sample mean  $\sum_{u \in S} y_u / 2$ . Is this an unbiased estimator?

$$E[\bar{y}] = 0.3 \left( \frac{1+3}{2} \right) + 0.5 \left( \frac{1+5}{2} \right) + 0.3 \left( \frac{3+5}{2} \right)$$

$$= 3.1$$

$$\mu = \frac{1+3+5}{3} = 3$$

$$\therefore \text{Sampling bias} = E[\bar{y}] - \mu = 3.1 - 3 = 0.1 \neq 0$$

- (b) [3 marks] Calculate the MSE of the sample mean.

$$MSE = \text{Var}(\bar{y}) + \text{Bias}(\bar{y})^2,$$

$$\text{where } \text{Var}(\bar{y}) = E[\bar{y}^2] - E[\bar{y}]^2$$

$$= 0.2(2^2) + 0.5(3^2) + 0.3(4^2) - 3.1^2 = 0.49$$

$$\text{So, } MSE = 0.49 + 0.1^2 = 0.5 //$$

6. [3 marks] State the three possible sources of error in a scientific study performed within the inductive inference framework.

Study error, sampling error, measurement error //

7. [6 marks (1 mark each)] Determine whether the following statements are true or false.

(a) The Horvitz-Thompson estimator is unbiased regardless of sampling design.

- i. True
- ii. False

(b) The observed significance level measures evidence in favour of the hypothesis  $H_0$ .

- i. True
- ii. False

(c) An unbiased estimator is always preferable to a biased estimator.

- i. True
- ii. False

(d) The exact value of an estimator's bias is calculable based on all possible samples.

- i. True
- ii. False

(e) The statistic  $D(\mathcal{P}_1, \mathcal{P}_2) = a(\mathcal{P}_1) - a(\mathcal{P}_2)$ , where  $a$  is an attribute, may or may not be a discrepancy measure.

- i. True
- ii. False

(f) Consider taking a random sample of size  $n < N$ , without replacement, from a population of size  $N$ . The equation  $P(S) = \pi_u$  is always true, where  $\pi_u$  is the probability of inclusion and  $P(S)$  is the probability of any sample of size  $n$ .

- i. True
- ii. False